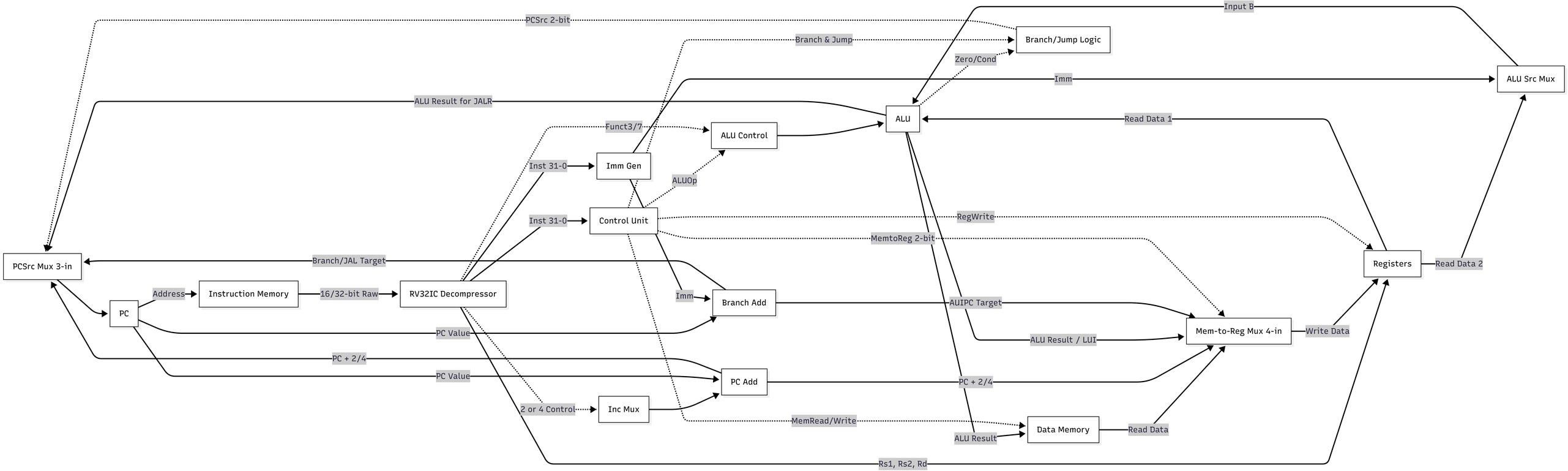
PCSrc 2-bit
Input B
Branch & Jump
Branch/Jump Logic
Zero/Cond
Imm
ALU Src Mux
ALU Result for JALR
ALU
Read Data 1
Funct3/7
ALU Control
Inst 31-0
Imm Gen
ALUOp
Inst 31-0
Control Unit
RegWrite
MemtoReg 2-bit
PCSrc Mux 3-in
Branch/JAL Target
Registers
Read Data 2
Address
Instruction Memory
16/32-bit Raw
RV32IC Decompressor
Imm
Branch Add
AUIPC Target
PC
PC Value
ALU Result / LUI
Mem-to-Reg Mux 4-in
Write Data
PC + 2/4
PC Value
PC Add
PC + 2/4
2 or 4 Control
Inc Mux
MemRead/Write
Data Memory
Read Data
ALU Result
Rs1, Rs2, Rd